

Food Inventory System for IUT Cafeteria

CSE 4408

Group : 10% SAD

Lab - 07

DATE : 29.07.2025



Project Context

An automated inventory system is required to track daily usage and manage stock levels efficiently

Briefly

- Manage inventory digitally
- Automate stock tracking
- Generate Financial and Stock alert



Prioritized User Stories

Role	Story	Priority
As a Cafeteria Manager	I want to view current inventory levels so that I can make informed purchasing decisions	Must
As a Cafeteria Manager	I want to generate summary reports so that I can monitor usage trends and wastage	Should
As a Cafeteria Manager	I want to Rerun Inventory Calculations to Improve Consistency and Keep Logs Orderly Long-term (R.I.C.K.R.O.L.L)	Could

Prioritized User Stories

Role	Story	Priority
As an Inventory Staff	I want to input daily usage of ingredients so that I can maintain accurate inventory records	Must
As an Inventory Staff	I want to get low-stock alerts so that I can notify upper management to restock on time	Should
As an IT Admin	I want to control access levels so that only authorized users can modify data	Must

Prototype

Cafeteria Inventory System Demo

Highlights : Role Based Login



Cafeteria Inventory
Sign in to manage your restaurant inventory

Select Role

☒ Manager

☐ Inventory Staff

Username
Enter your username

Password
Enter your password

Sign In



Cafeteria Inventory
Sign in to manage your restaurant inventory

Select Role

☐ Manager

☒ Inventory Staff

Username
staff1

Password
...

Sign In



Cafeteria Inventory
Sign in to manage your restaurant inventory

Select Role

☒ Manager

☐ Inventory Staff

Username
manager1

Password
...

Sign In

Prototype

Cafeteria Inventory System Demo

Highlights : Inventory Staff Login and Dashboard



Cafeteria Inventory

Sign in to manage your restaurant inventory

Select Role

☐ Manager

☒ Inventory Staff

Username

Password



Sign In

 Inventory Dashboard Staff View		+ Record Usage	 4	Logout
Search inventory items...		All Items	Low Stock	In Stock
 Inventory Items (8)				
Item Name	Current Stock	Threshold	Status	
Lettuce	15 heads	20 heads	Low Stock	
Tomatoes	45 kg	30 kg	In Stock	
Chicken Breast	8 kg	15 kg	Low Stock	
Rice	100 kg	25 kg	In Stock	

Prototype

Cafeteria Inventory System Demo

Highlights : In Stock and Low Stock pages

Warning: 4 items are below stock threshold!

Inventory Dashboard

Staff View

+ Record Usage

4

Logout

Search inventory items...

All Items

Low Stock

In Stock

Inventory Items (4)

Item Name	Current Stock	Threshold	Status
Tomatoes	45 kg	30 kg	In Stock
Rice	100 kg	25 kg	In Stock
Onions	80 kg	20 kg	In Stock
Milk	50 liters	15 liters	In Stock

Warning: 4 items are below stock threshold!

Inventory Dashboard

Staff View

+ Record Usage

4

Logout

Search inventory items...

All Items

Low Stock

In Stock

Inventory Items (4)

Item Name	Current Stock	Threshold	Status
Lettuce	15 heads	20 heads	Low Stock
Chicken Breast	8 kg	15 kg	Low Stock
Olive Oil	3 bottles	10 bottles	Low Stock
Bread	25 loaves	30 loaves	Low Stock

Prototype

Cafeteria Inventory System Demo

Highlights : Stock Update board for verified Inventory Staff

Record Item Usage

Select Item

Choose an inventory item

▼

Quantity Used

Enter quantity

📅

Date

07/28/2025

📅

📁

Record Usage

Record Item Usage

Select Item

Choose an inventory item

▼

Lettuce (15 heads available)

Tomatoes (45 kg available)

Chicken Breast (8 kg available)

Rice (100 kg available)

Olive Oil (3 bottles available)

Onions (80 kg available)

📁

Record Usage

Usage Recorded

20 kg of Rice recorded for 2025-07-28

Prototype

Cafeteria Inventory System Demo

Highlights : Report and Summary Dash board for Cafeteria Manager



Reports & Analytics

Generate inventory reports and insights

[Back to Dashboard](#)

Generate Report

Start Date

07/21/2025



End Date

07/28/2025



Generate Report



Select Date Range

Choose your date range and click "Generate Report" to view inventory analytics and insights.

Prototype

Cafeteria Inventory System Demo

Highlights : Timeline wise analytics and trend

07/21/2025

07/28/2025

Generate Report

Download PDF

25
Total Items

4
Low Stock

\$15,420.5
Total Value

12.5%
Usage Increase

Usage Report (2025-07-21 to 2025-07-28)

Item	Quantity Used	Date
Lettuce	25 heads	1/20/2024
Tomatoes	12.5 kg	1/20/2024
Chicken Breast	8.2 kg	1/19/2024
Rice	15 kg	1/19/2024
Olive Oil	2 bottles	1/19/2024

Low Stock Items

Item	Current Stock	Threshold	Priority
Lettuce	15	20	Medium Priority
Chicken Breast	8	15	High Priority
Olive Oil	3	10	High Priority
Bread	25	30	Low Priority

Warning: 4 items are below stock threshold!

Prototype

Cafeteria Inventory System Demo

Tool Used: Lovable, Basic HTML, CSS

Type: Selected Features UI/UX Prototype

Justification:

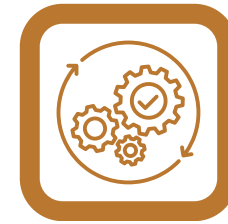
- Early visualization before backend development
- Validates design with stakeholders
- Navigational prototype for better initial experience

Scrum Roles



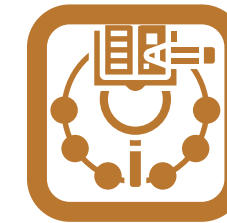
Product Owner: The Visionary

- Project Creator or the ship's captain
- Led stakeholder interviews and gathered key requirements.
- Knows user needs and prioritizes features effectively.
- Maintains the Product Backlog and maximizes business value.



Scrum Master : The Coach

- Organized and coordinated previous lab work.
- In charge of Scrum events and ensures smooth team communication.
- Keeps the team focused on goals.



Development Team: The Builders

- Design database and application logic.
- Implements backend functionality and ensures data accuracy.
- Will seek help from a friend or two
- Partner up with IUT IT personnel for database integration

Sprint Planning

Sprint 01

Must, Medium

Inventory usage input screen

Must, Large

Role-Based User authentication

Should, Small

Basic alert system for low stock

Duration : 2 Weeks

Rationale: These are foundational, lower complexity, and unlock later modules

Thanks!

Do you have any questions?
Feel free to ask now

Contributors



Shadab
220041201



Zia
220041245



Lihan
220041203

